

AI Chatbot Specification

■ F&B; / Restaurants

AI chatbot specification for restaurants, cafes, hawker stalls, and catering businesses in Singapore.

Suggested Pricing

Component	Price
One-time setup fee	\$300 - \$500
Monthly maintenance	\$60 - \$100/month
Add-on: WhatsApp integration	+\$150-200
Add-on: Payment collection	+\$100-150
Add-on: Custom integrations	+\$100-200 each

Business Pain Points (What You Sell Against)

These are the problems the business owner already knows they have. Lead with these in your pitch — don't sell 'AI', sell the solution to their pain.

- Staff spend 1-2 hours daily answering calls and WhatsApp messages about menu, hours, and reservations.
- Catering enquiries require long back-and-forth to gather event details, pax, dietary requirements.
- Customers leave when they can't get a quick answer about dietary options (halal, vegetarian, gluten-free).
- Reservation no-shows waste tables and revenue — no deposit system in place.
- Social media DMs (Instagram) go unanswered for hours during peak service.
- Google reviews mention 'hard to contact' or 'slow response' — hurting online reputation.

30-Second Sales Pitch

I'll build an AI assistant for your restaurant that handles menu questions, takes reservations, and manages catering enquiries — on your website, WhatsApp, and Instagram. Customers get instant answers, you stop losing reservations because no one picked up the phone. Setup takes 3 days.

Feature Requirements

Build in order: Must Have first (your MVP), then Should Have (first upsell), then Nice to Have (premium tier).

Feature	Description	Priority	Effort
Menu & Pricing FAQ	Answer questions about dishes, prices, ingredients, dietary options	Must Have	Low
Operating Hours & Location	Business hours, address, directions, parking info	Must Have	Low
Table Reservation	Book a table by collecting date, time, pax, and contact	Must Have	Medium
Catering Enquiry	Collect event details for catering quotes	Should Have	Low
Promotions & Loyalty	Show current promotions, happy hour deals, loyalty programmes	Nice to Have	Low
Delivery/Takeaway Orders	Take simple orders for pickup or redirect to delivery platforms	Nice to Have	High

Detailed Feature Specifications

Use these specifications when building with Claude Code. Each feature includes the user story, technical implementation notes, and acceptance criteria.

Feature: Menu & Pricing FAQ

Priority: Must Have | **Effort:** Low

User Story: As a customer, I want to ask 'do you have vegetarian options under \$15' and get specific dish recommendations.

Technical Notes: Upload menu PDF. RAG retrieves relevant items. Claude formats a helpful response with dish name, price, and brief description. Handle dietary filter queries: halal, vegetarian, vegan, gluten-free, nut-free.

Acceptance Criteria:

- Answers pricing questions accurately
- Filters by dietary requirement correctly
- Handles 'what do you recommend' with top 3 suggestions
- Mentions allergen info when asked

Feature: Operating Hours & Location

Priority: Must Have | **Effort:** Low

User Story: As a customer, I want to know if the restaurant is open right now and how to get there.

Technical Notes: Store hours as structured data with public holiday overrides. Include Google Maps link. Mention nearest MRT and bus stops. Handle 'are you open now' with real-time check against current SG time (UTC+8).

Acceptance Criteria:

- Correct hours for each day including different weekend hours
- Handles public holidays correctly
- Provides Google Maps link and nearest MRT
- 'Are you open now' checks against current time

Feature: Table Reservation

Priority: Must Have | **Effort:** Medium

User Story: As a customer, I want to book a table for Saturday dinner without calling the restaurant.

Technical Notes: Collect: date, time, pax, name, phone. Check availability against Google Calendar or reservation system. Cap pax at max table size. Send confirmation with address. For large groups (8+), route to human.

Acceptance Criteria:

- Collects all required info conversationally (not as a form)

- Validates date is in the future and within opening hours
- Handles 'tonight', 'tomorrow', 'this Saturday' correctly
- Sends confirmation with restaurant address
- Routes groups 8+ to human for special arrangements

Feature: Catering Enquiry

Priority: Should Have | **Effort:** Low

User Story: As an event organiser, I want to submit my catering requirements and get a quote without multiple emails.

Technical Notes: Conversational form: event type, date, pax, budget range, dietary needs, location. Compile into a structured email sent to the catering manager. Confirm to customer that someone will respond within 24hrs.

Acceptance Criteria:

- Collects: event type, date, pax, budget, dietary, location
- Sends formatted email to catering manager
- Gives customer a reference number
- Provides ballpark pricing from menu if available

Feature: Promotions & Loyalty

Priority: Nice to Have | **Effort:** Low

User Story: As a customer, I want to know about current promotions or deals before I visit.

Technical Notes: Simple CMS or Google Sheet with active promotions. Bot checks and returns current deals. Support time-based promos (happy hour 5-7pm). Admin can update promos without code changes.

Acceptance Criteria:

- Shows currently active promotions only (not expired ones)
- Handles 'any discount', 'got promo', 'happy hour' queries
- Admin can add/remove promos via simple interface

Feature: Delivery/Takeaway Orders

Priority: Nice to Have | **Effort:** High

User Story: As a customer, I want to order food for pickup or be directed to the right delivery app.

Technical Notes: For MVP: redirect to GrabFood/FoodPanda links. For advanced: collect order items, calculate total, accept PayNow payment, send order to kitchen display/Telegram. Start with redirect, upgrade later.

Acceptance Criteria:

- Provides correct links to delivery platforms
- For takeaway: collects order, calculates total, provides pickup time estimate

- Sends order to kitchen via Telegram notification

Sample Conversations

These demonstrate expected chatbot behaviour. Use these as test cases during development.

Scenario: Customer asking about menu and making a reservation

Customer: Got any gluten free options?

Bot: Yes! We have 6 gluten-free dishes: Grilled Salmon (\$18), Thai Basil Chicken with rice (\$14), Green Curry with rice (\$15), Caesar Salad (no croutons, \$12), Mango Sticky Rice (\$8), and Coconut Panna Cotta (\$9). Would you like to make a reservation?

Customer: Can book table for Saturday 7pm, 4 pax

Bot: Let me check... Saturday 7pm for 4 is available! Booked under what name and contact number?

Customer: Sarah, 8765 4321

Bot: All set! Reservation confirmed: Saturday 7pm, 4 pax, under Sarah. We're at 123 Orchard Road, #01-05 (nearest MRT: Somerset). See you Saturday!

Scenario: Catering enquiry

Customer: I need catering for an office event

Bot: I'd be happy to help! Let me get some details. How many people are you catering for, and what's the date?

Customer: About 30 people, next Friday lunch

Bot: Got it. Do you have a budget in mind, and any dietary requirements (halal, vegetarian, allergies)?

Customer: Budget around \$20-25 per pax, need some vegetarian options

Bot: Perfect. I've sent your enquiry to our catering manager: 30 pax, next Friday lunch, \$20-25/pax, vegetarian options needed. They'll get back to you within 24 hours with a detailed menu proposal. Your reference number is CAT-20260505. Is there anything else?

Technical Architecture

Core Stack (Required)

Component	Technology	Purpose
LLM	Claude API (claude-sonnet-4-20250514)	Understands questions and generates answers from context
Vector Database	ChromaDB (local) or Pinecone (cloud)	Stores document embeddings for RAG retrieval
Embedding Model	Voyage AI or OpenAI text-embedding-3-small	Converts text chunks to vectors

Backend	Python FastAPI	API server handling chat requests
Frontend Widget	Vanilla JS embeddable snippet	Chat bubble on client's website
Admin Dashboard	Streamlit or Next.js	Where business owner uploads docs and views analytics

Add-on Integrations

- **Google Calendar API:** Reservation management and availability checking
- **WhatsApp Business API:** Primary channel for SG F&B; customers
- **Instagram DM API (Meta):** Auto-reply to Instagram enquiries
- **Telegram Bot:** Kitchen notifications for takeaway orders, admin alerts
- **Stripe / PayNow:** Deposit for large group reservations, takeaway payment
- **Google Sheets:** Simple menu management, promotion updates

Data Model (Documents to Collect from Client)

These are the documents and data you need from the business owner during onboarding. The quality of these documents directly determines chatbot accuracy.

- **Full Menu:** PDF or spreadsheet: dish name, price, description, ingredients, dietary tags (halal, vegetarian, GF, nut-free)
- **Operating Hours:** Regular hours + public holiday schedule + any seasonal changes
- **Location Info:** Full address, Google Maps link, nearest MRT/bus, parking options
- **Catering Menu:** Catering packages, per-pax pricing, minimum order, lead time required
- **Reservation Policy:** Max pax per booking, cancellation policy, deposit requirements for large groups
- **Current Promotions:** Active deals, happy hour timing, loyalty programme details

PDPA Compliance Checklist (Singapore)

- Display clear data collection notice before collecting any personal data
- Get explicit consent before storing name, phone, email
- Provide opt-out option: 'Type STOP to opt out of data collection'
- Do not store NRIC, credit card numbers, or medical records in chat logs
- Implement data retention policy: auto-delete chat logs after 90 days
- Provide data export/deletion on request from the business owner
- Add privacy policy link in the chat widget footer

Singapore Localisation Notes

- Support Singlish phrasing: 'got discount ah?', 'how much one?', 'can book or not'
- Handle mixed language queries (English + Mandarin terms are common)
- Use SGD (\$) for all pricing — never USD
- Time format: 12-hour with AM/PM (e.g., 3pm, not 15:00)
- Support PayNow QR code generation for payments
- Reference local landmarks for directions ('near Jurong East MRT')
- Public holiday awareness for Singapore calendar